Computer Vision and CNN(suggestion)

SECTION A: Short Answer Questions (2 Marks Each)

Q1. Define an image feature and give one real-world example.

An image feature is a distinctive visual characteristic of an image that a computer can identify and use for analysis.

Example: A face recognition system uses facial features such as the distance between the eyes to identify a person.

Q2. Differentiate between edge features and texture features.

- Edge features: Identify boundaries where image intensity or colour changes sharply. Example: the outline of a door.
- Texture features: Describe repeated patterns or surface details. Example: the rough texture of a brick wall.

Q3. Which feature detects the rectangular outline of a smartboard, and why?

Edge features are used because they detect sharp changes in brightness or colour along the smartboard's rectangular boundaries.

Q4. What is a corner feature?

A corner feature is a point in an image where two or more edges meet or where the image intensity changes significantly in multiple directions.

Example: The corner of a window where its horizontal and vertical edges intersect.

Q5. How do colour features distinguish a green football field from a grey running track?

Colour features analyse the colour values of pixels. Green pixels are associated with the football field, while grey pixels are associated with the running track, allowing the computer to distinguish between them.

Q6. Why is feature extraction more efficient than treating every pixel equally?

Feature extraction focuses on important information, such as edges, shapes and textures, instead of processing every pixel as equally meaningful. This can reduce computational effort and simplify image analysis.

Q7. State two practical applications of image features.

1. Face recognition in security systems.
2. Object detection in autonomous vehicles.

Q8. Define convolution in a CNN.

Convolution is a mathematical operation in which a small filter (kernel) slides across an input image and performs element-wise multiplication and addition to extract visual patterns.

Q9. What is a kernel or filter in image processing?

A kernel is a small matrix of numerical values that moves across an image to detect specific patterns, such as edges, corners or textures.

Example: A 3×3 kernel used for edge detection.

Q10. Calculate the output size.

Given:

- Input size = 28×28
- Kernel size = 5×5
- Stride = 1
- Padding = 0

Formula:

$$Outputsize = \frac{N-F+2P}{S} + 1$$

$$= \frac{28-5+0}{1} + 1 = 24$$

Answer: 24×24.

Q11. Explain the role of stride.

Stride specifies the number of pixels the kernel moves at each step.

- Stride 1: moves one pixel at a time.

- Stride 2: moves two pixels at a time.

A larger stride generally produces a smaller feature map.

Q12. How are kernels created in traditional image processing versus CNNs?

- Traditional image processing: Kernel values are manually designed by humans for a specific task, such as edge detection.
- CNN: Kernel values are learned and adjusted automatically during training using labelled data.

Q13. How does a CNN adjust kernel values during training?

A CNN compares its predictions with the correct labels to calculate a loss. Backpropagation calculates gradients, and an optimisation algorithm such as gradient descent updates the kernel values to reduce the loss.

Q14. Define a feature map.

A feature map is the output produced when a kernel is applied to an input image through convolution. It indicates where and how strongly a particular pattern is detected.

Q15. Why is pooling used after convolution?

Pooling reduces the spatial dimensions of a feature map, decreasing computation and memory requirements while helping retain important visual information.

Q16. Is pooling another form of learning a kernel?

Incorrect. Pooling is a fixed operation, such as maximum or average pooling. Unlike convolutional kernels, its values are not normally learned during training.

Q17. Write the full form of ReLU and its formula.

ReLU stands for Rectified Linear Unit.

Its mathematical formula is:

$$f(x) = \max(0, x)$$

Q18. State two reasons why ReLU is widely used.

1. It introduces non-linearity, enabling neural networks to learn complex patterns.
2. It is computationally simple and helps reduce the vanishing-gradient problem compared with sigmoid or tanh in many settings.

Q19. How does ReLU treat negative numbers?

ReLU converts negative values to zero while leaving positive values unchanged.

For example:

$$f(-4) = 0, f(6) = 6$$

Q20. What is a No-Code AI tool, and who is its target audience?

A No-Code AI tool allows users to create or train AI models through graphical interfaces without writing programming code.

Its primary audience includes students, teachers, beginners and non-programmers.

Q21. What is the main objective of a convolutional layer?

The main objective is to extract meaningful visual patterns from input images, such as edges, shapes and textures, using learnable filters.

Q22. List three low-level features extracted by the first CNN layer.

1. Edges.
2. Corners.
3. Simple textures.

Q23. What is Teachable Machine?

Teachable Machine is a browser-based tool by Google that allows beginners to train simple machine learning models using images, sounds or poses without programming.

Users provide examples, train a model and test its predictions.

Q24. Define Artificial Neural Networks based on their ability to handle raw data.

An Artificial Neural Network (ANN) is a machine learning model made up of interconnected artificial neurons that learns patterns and relationships from

data. Depending on its architecture, it can process raw data or data that has already been converted into numerical features.

Q25. Name an open-source data visualisation and analysis tool used for data mining and ML.

Orange is an open-source data visualisation, analysis and machine learning tool that uses a visual, widget-based workflow.

SECTION B: Long Answer Questions (4 Marks Each)

Q26. Smart attendance system using facial recognition

1. Use of shape and colour features

- Shape features: The system identifies the facial outline, eyes, nose and mouth to distinguish a face from the classroom wall.
- Colour features: It analyses differences in colour between skin, hair, clothing and the background to help separate a student from surrounding objects.

Modern face recognition systems generally combine these visual cues with learned features from a neural network.

2. Importance of object recognition

Object recognition helps the system locate and identify students' faces in a crowded classroom. Once a face is detected, the system can compare its learned facial representation with registered student data and record attendance for a recognised student.

Q27. CNN convolution for drone landing-pad detection

1. Calculate the output dimensions

Given:

- Input image: 10×10
- Kernel: 3×3
- Stride: 1
- Padding: 0

Using the formula:

$$O = \frac{N-F+2P}{S} + 1$$

$$O = \frac{10-3+0}{1} + 1 = 8$$

Answer: The output feature map is 8×8.

2. Step-by-step convolution process

1. Place the 3×3 kernel over the top-left 3×3 region of the image.
2. Multiply each kernel value by the corresponding image pixel value.
3. Add all nine products to obtain one output value.
4. Move the kernel one pixel to the right and repeat the calculation.
5. After reaching the right edge, move down one row and continue.
6. Repeat until all valid positions have been processed.

The kernel produces 8 positions horizontally and 8 vertically, creating a feature map containing $8 \times 8 = 64$ values.

Q28. ReLU in a smartphone face-unlock system

1. Why is a non-linear activation function needed?

Convolution performs linear operations. If a neural network used only linear transformations, stacking multiple layers would still be equivalent to a single linear transformation (ignoring other operations). This would limit its ability to learn complex facial patterns.

ReLU introduces non-linearity by applying:

$$f(x) = \max(0, x)$$

This helps the network learn complex combinations of facial features.

2. Does ReLU automatically delete all unimportant features?

No. ReLU sets negative activation values to zero and keeps non-negative values unchanged. It does not understand whether a feature is important to face recognition.

Other network components and the training process determine which patterns are useful for identifying a face.

Q29. No-code AI project for sorting apples

1. Suggested platform: Google Teachable Machine

The student can use Teachable Machine to create a simple image classification model.

Steps:

1. Create two classes: Ripe Apples and Unripe Apples.
2. Capture or upload several images of red and green apples using a webcam.
3. Train the model using the provided examples.
4. Test the model with new apple images.
5. Use the predictions to classify apples into the two categories.

No programming is required to train and test the basic model.

2. Two benefits of no-code AI platforms

- Easy to use: Students can develop AI projects without learning a programming language first.
- Interactive learning: Students can observe how training examples influence predictions and understand basic machine learning concepts through experimentation.

Q30. Pooling explained using a tourist map

1. Mapping the scenario to pooling

When studying a city map, a person ignores tiny lanes and focuses on major landmarks. Similarly, pooling reduces the spatial size of a CNN feature map while retaining representative information.

For example, max pooling over a 2×2 region selects the largest value from those four values.

2. Why is reducing spatial size helpful?

- Reduces the number of values processed by later layers.
- Reduces memory usage and computational cost.
- Can make the representation less sensitive to small shifts in object position.
- Helps later layers focus on stronger, more relevant patterns.

However, pooling can also remove fine details, so it must be used appropriately.

Q31. Traditional edge detection versus CNNs in self-driving cars

1. How are kernel values obtained?

- Situation A — Traditional software: A programmer manually selects kernel values based on mathematical knowledge of edge detection. For example, a Sobel filter is designed to detect intensity changes in a particular direction.
- Situation B — CNN: The kernels are generally initialised with numerical values and adjusted automatically during training using labelled images, backpropagation and an optimisation algorithm.

2. Why is the CNN method advantageous?

Road conditions vary across cities due to lighting, weather, camera angles, sign designs and backgrounds. CNNs can learn useful patterns from diverse training examples instead of relying only on manually designed filters.

This adaptability can improve recognition in varied environments, provided the training data is sufficiently diverse. A CNN still requires careful testing and validation for safe autonomous driving.

Q32. Agricultural robot detecting leaf diseases

1. Which layer extracts low-level features?

The first convolutional layer typically learns low-level visual features, such as edges, simple textures and the jagged boundary of a leaf.

2. Workflow through the CNN

Step 1: Convolution

The filters slide across the leaf image and detect patterns such as edges, veins and spots, producing feature maps.

Step 2: ReLU

Applies $f(x) = \max(0, x)$, setting negative activations to zero and introducing non-linearity.

Step 3: Pooling

Reduces the feature map's spatial dimensions while retaining representative information.

These operations may be repeated across multiple layers before the model predicts whether the leaf is healthy or diseased.

Q33. Detecting bone fractures in X-ray images

1. Importance of edge features

Edge features identify sudden changes in image intensity. In X-rays, a fracture may appear as a thin line, a break in the bone's normal outline or a disruption in its internal structure.

Edge-sensitive filters can help the network identify these changes and distinguish them from the surrounding bone. Very fine fractures may require high-resolution images and multiple learned features.

2. What happens if ReLU is omitted?

Without non-linear activation functions, a network made only of linear convolutional transformations would remain effectively linear, even with multiple layers. It would struggle to represent complex bone structures and subtle fracture patterns.

ReLU enables the network to learn more complex relationships between visual features. Other non-linear operations can serve a similar purpose.

Q34. Feature maps for detecting vertical lines on book covers

1. How is a feature map generated?

A vertical-line filter is placed over the image of the books. At each position, the filter values are multiplied by the corresponding image values and summed.

The resulting number is stored at the matching position in the output feature map. The kernel then moves across the image, repeating the process.

2. What does each numerical value represent?

Each value indicates how strongly the local image region matches the pattern detected by the filter.

- A strong response may indicate a vertical line.
- A weak response may indicate little evidence of that pattern.
- A negative response may indicate a contrasting pattern, depending on the filter.

The robot can use this information, alongside other features, to analyse book-cover patterns and help identify incorrectly shelved books.

Q35. CNN versus ANN and Orange

1. Define CNN and distinguish it from a basic ANN.

An Artificial Neural Network (ANN) is a broad family of models that learn relationships between input data and output predictions.

A Convolutional Neural Network (CNN) is a specialised type of neural network that uses convolutional filters to learn spatial patterns in data, especially images.

ANN (basic fully connected network)	CNN
Often works with fixed-length numerical feature vectors.	Designed to exploit spatial relationships in images.
Connects neurons across successive layers.	Uses local filters and shared weights.
May require manual feature extraction for complex images.	Can learn visual features directly from image pixels.

2. Why use Orange for numeric data and CNNs for photographs?

- Orange: Provides a visual, drag-and-drop workflow for data mining, statistical analysis, data visualisation and machine learning on tabular data, such as student marks or sales figures.
- CNN: Is particularly suitable for photographs because convolution learns local spatial patterns such as edges, textures and shapes.

A CNN is not strictly limited to photographs, and Orange is not limited to numeric data. The best tool depends on the data type, task and model requirements.

Quick revision: Important formulas and concepts

Topic	Key point
Convolution output	\(\displaystyle O=\frac{N-F+2P}{S}+1\)
ReLU	$f(x) = \max(0, x)$
Feature map	Output generated by applying a filter to an input
Stride	Number of pixels the filter moves per step
Pooling	Reduces spatial dimensions

CNN training	Learns filter values using backpropagation and optimisation
Low-level features	Edges, corners and simple textures
No-code AI	Builds models through graphical tools without requiring code